Wage Inequality in Turkey

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Presentation Outline

1. **Motivation and background**
   Why wage inequality matters, and why we study it in the Turkish context.
2. **Data and methodology**
   Overview of HBS and SILC datasets, key variables, and empirical strategy.
3. **Measuring wage inequality**
   Gini coefficient, variance of log wages, percentile ratios, and wage premiums by gender, education, experience.
4. **Results**
   Our estimations and findings via SILC(2023) and HBS(2022).
5. **Comparison with the World**
   Comparing Turkey with Other Countries in terms of wage inequality levels.
6. **Review and Conclusions**
   Review of the literature, comparing to our findings and conclusions.

What is Wage Inequality?

- Wage inequality refers to differences in labor earnings across individuals in the workforce.
- Unlike income inequality, it excludes capital income, transfers, or wealth—it focuses only on wages and salaries.
- It reflects labor market dynamics such as education, experience, gender, region, and institutional settings (e.g., minimum wage, unionization).
- It can exist within a company, sector, or across the whole economy.

Wage vs Income vs Wealth Inequality

	What it Measures	Main Components
Wage Inequality	Differences in salaries/wages among workers	Hourly/monthly earnings from labor
Income Inequality	Differences in total income across individuals	Wages, self-employment income, capital income, transfers
Wealth Inequality	Differences in assets and net worth	Property, savings, stocks, inheritance

*Wage inequality is a part of income inequality.
*Wealth inequality is usually more extreme and persistent over generations.

How Do We Measure Wage Inequality?

Gini Coefficient

- Measures overall wage inequality (0 = perfect equality, 1 = complete inequality).

Percentile Ratios (e.g. P90/P10, P50/P10)

- Compares high and low earners (e.g., wage at 90th percentile vs. 10th).

Share of Workers Below Certain Thresholds

- Example: % of workers earning less than 1.05× minimum wage (Eurostat, 2018: ∼42% in Turkey).

Standard Deviation or Variance of Wages

Minumum Wage

- Share of wage earners and casual employees earning less than 1.05 times the minimum wage in total wage earners and casual employees (percentage, 2018)

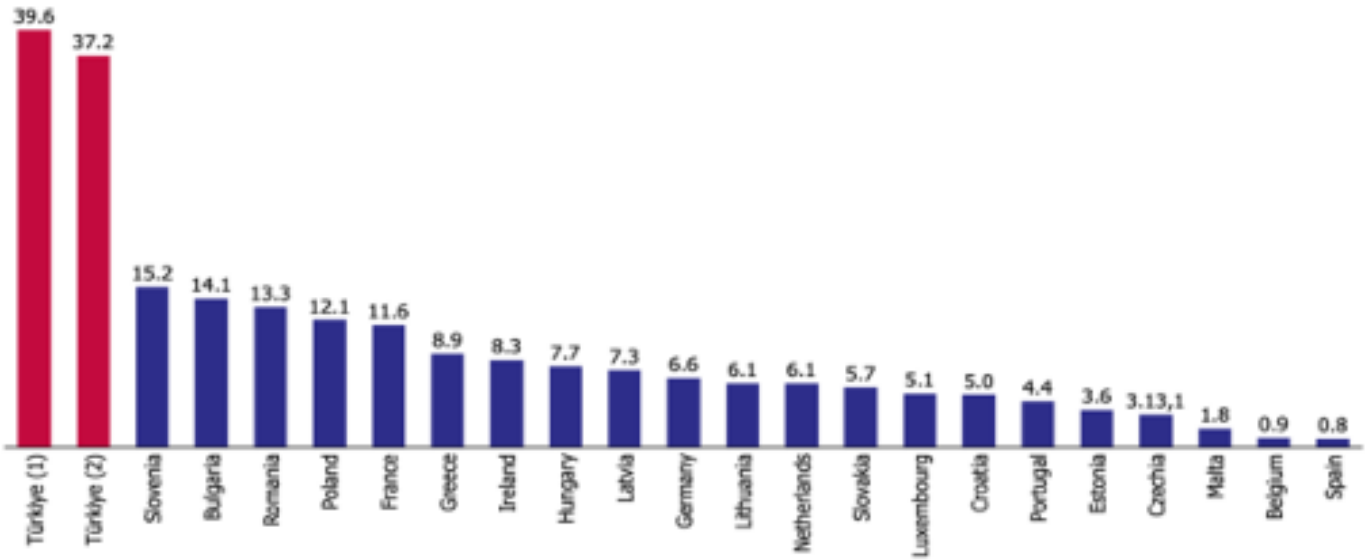


Figure: Cunedioğlu, H. E. (2023)

What Causes Wage Inequality?

Factor	Impact
Human Capital	Education, skills, experience gaps
Technological Change	Higher demand for skilled workers
Labor Market Structure	Informality, weak unions
Globalization	Outsourcing pressures

- These factors interact and reinforce inequality

Who Suffers the Most?

- Low-educated workers: Low-paying, insecure jobs
- Women: Persistent gender wage gap
- Young people: High unemployment, entry-level gaps
- Underdeveloped regions: East/Southeast Turkey

Structural Disadvantage

These groups face systemic labor market barriers

Why Should We Care?

1. Limits economic growth (low spending power)
2. Reduces social mobility
3. Increases social tension
4. Weakens public trust
5. Creates intergenerational cycles

What Do Researchers Say?

Key Thinkers

- Piketty: Capital accumulation increases inequality
- Atkinson: Need education and taxation policies
- Chetty: Inequality lowers mobility
- Torul: Turkey's persistent education-informality link

Common Misunderstandings

"Work harder $\rightarrow$ earn more" Opportunity matters

"Inequality motivates" Can demotivate instead

"Natural market outcome" Policy shapes inequality

"Only affects poor" Harms whole society

What Wage Inequality Does Not Tell Us

A high wage inequality index does not mean a country is poor or weak.

- United States example: High wage inequality but also high GDP per capita
- Low wage inequality economic fairness or prosperity
- Many factors affect inequality: automation, policy changes, migration
- Not a standalone measure of economic health or global power

Data: Household Budget Survey (HBS)

- **Coverage:**
  - Disposable income and consumption subcomponents (household level)
  - Labor supply, income sources (individual level):
    - Labor earnings
    - Asset income
    - Self-employment income
    - Government/private transfers
  - We focused on labor earnings and self-employment income.
- **Sample:**
  - 39,189 person (2022)
  - Nationally representative

Data: Survey of Income and Living Conditions (SILC)

- **Coverage:**
  - Detailed income sources for all household members
  - We focused on labor earnings and self-employment income
- **Sample:**
  - 60,223 person (2023)
  - Nationally representative
- **Suitability:**
  - Both HBS and SILC comply with RED guidelines for inequality studies
    Source: Tamkoç and Torul (2020)

Estimation Method

- **Note:**
  - SILC (2023) and HBS (2022) are used for estimation.
- **Data Adjustments:**
  - All monetary variables converted to real terms using CPI (2002 base).
- **Sample Restrictions:**
  - Working-age individuals (25-60 years).
  - Exclusions:
    - Non-reporting of hours worked.
    - Annual earnings $<$ 2439 TL.

Note on Threshold

2439 TL in 2022 = ˜50% of Turkey's minimum wage at the time.

To be compatible with RE

Annual Earnings Calculation

$$a_{e,i,t} = nw_{i,t} + rw_{i,t} + p_{i,t} + b_{i,t} + \alpha_{\text{TUR}}(nse_{i,t} + rse_{i,t} + ag_{i,t})$$

- **Components:**

Symbol	Description
$nw_{i,t}$	Annual cash payments
$rw_{i,t}$	Annual real payments
$p_{i,t}$	Annual premiums
$b_{i,t}$	Annual bonuses
α_{TUR}	Labor share of national income $\sim 42\%$ [1]
$nse_{i,t}$	Cash self-employment income
$rse_{i,t}$	Real self-employment income
$ag_{i,t}$	Net agricultural income

[1] As of 2019, it is calculated as $\approx 43\%$ with decreasing trend.
Source: https://fred.stlouisfed.org/series/LABSHPTRA156NRUG

Wage and Premium Calculations

- **Hourly Wage:**

$$w_{i,t} = \frac{ae_{i,t}}{ah_{i,t}} \quad \text{(Annual earnings / annual hours worked)}$$

- **Premium Definitions:**
  - Gender:

$$\text{Gender Premium}_t = \frac{\overline{w_{m,t}}}{\overline{w_{f,t}}}$$

    (Male vs. female avg. wages)
  - Education:

$$\text{Education Premium}_t = \frac{\overline{w_{c,t}}}{\overline{w_{h,t}}}$$

    (College vs. HS grad males)
  - Experience:

$$\text{Experience Premium}_t = \frac{\overline{w_{o,t}}}{\overline{w_{y,t}}}$$

    (Older vs. younger males)

Inequality Measurement

- **Four Key Metrics:**
- Variance of Residuals
- Gini Coefficient
- Percentile ratios (P90/P50, P50/P10)
- Variance Log Wage

Inequality Measurement(Cont'd)

Variance of Residuals:

$$\log(y_{i,t}) = c_t + \beta_1 D^s_{i,t} + \beta_2 D^e_{i,t} + \beta_3 D^f_{i,t} + \beta_4 D^a_{i,t} + \beta_5 D^{ce}_{i,t} + \epsilon_{i,t}$$

Then, calculate $\hat{\epsilon_{i,t}}$, residual error term.

- **Control Variables:**

Symbol	Description
D^s	Gender dummy
D^e	11 education dummies
D^f	Family composition dummies
D^a	7 age category dummies
D^{ce}	4 couple education dummies

Inequality Measurement (Cont'd) – Gini Coefficient

Gini Coefficient

$$G = \frac{A}{A + B}$$

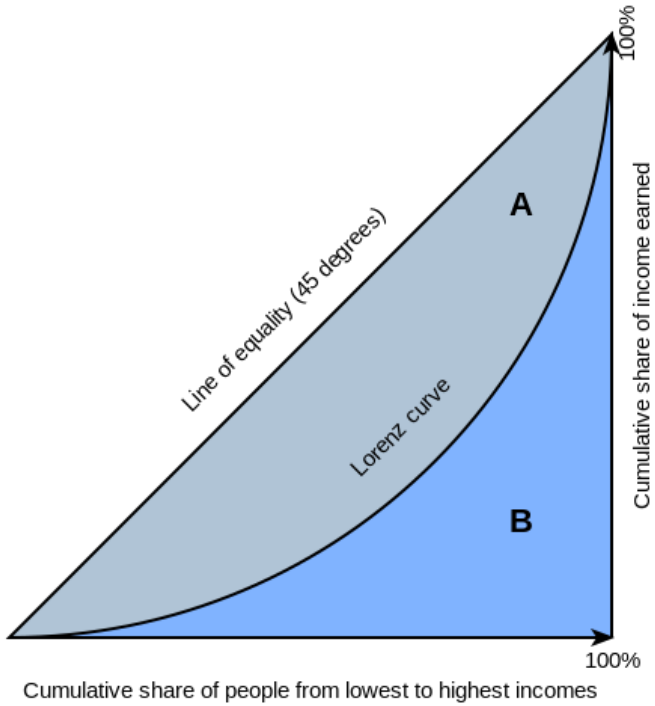


Figure: Lorenz curve and the line of absolute equality

Inequality Measurement (Cont'd)

Percentile Ratios

- **P90/P50**: Top vs. median earnings
- **P50/P10**: Median vs. bottom earnings

Variance of Log Wage

$$\text{Var}(\log W) = \frac{1}{N}\sum_{i=1}^{N}(\log w_i - \overline{\log w})^2$$

- Measures proportional wage differences
- Log transform reduces skewness

How Wage Earners Decided

- For calculating wage inequality, RED guidelines were used.
- Working age individuals counted (aged 25-60).
- Excluded individuals do not report hours worked.
- Excluded individuals whose annual earnings are below 2,439 Turkish Liras (i.e half of Turkey's monthly minimum wage in 2022).

Descriptive Statistics About Hourly Wage (2022)

	Min	Max	Mean	Median	Std. Dev.
SILC - Overall	1.3846	1079	42.7545	31.7307	44.1467
SILC - Male	1.3846	1079	43.2685	31.7692	46.4847
SILC - Female	1.4690	807	41.3580	31.1965	37.0245
HBS - Overall	5.8080	633	136.8628	115.9861	99.4841
HBS - Male	5.8500	633	137.4669	116.6666	98.3245
HBS - Female	5.8080	625	135.1767	112.5000	102.6521

All values are TL — 1% Threshold is removed from HBS to remove outliers

Inequality and Wage Premium Indicators (2022)

	Gini	Var(Log Wages)	Edu. Premium	Gender Premium	Exp. Premium	P90/P50	P50/P10	Var(Residuals)
SILC	0.4120	0.6304	1.6463	1.0461	1.0509	2.5454	2.8448	0.4605
HBS	0.4304	0.6983	1.5623	1.0050	0.9175	2.3535	2.4871	0.5058

Table: Our estimations for 2022

	Gini	Var(Log Wages)	Edu. Premium	Gender Premium	Exp. Premium	P90/P50	P50/P10	Var(Residuals)
SILC	0.4241	0.6580	1.5441	1.0265	1.0037	2.5013	2.7165	0.4898
HBS	0.4173	0.6480	1.5884	1.0095	0.9046	2.3413	2.8752	0.5902

Table: Torul's estimations for 2022

Regression Specification

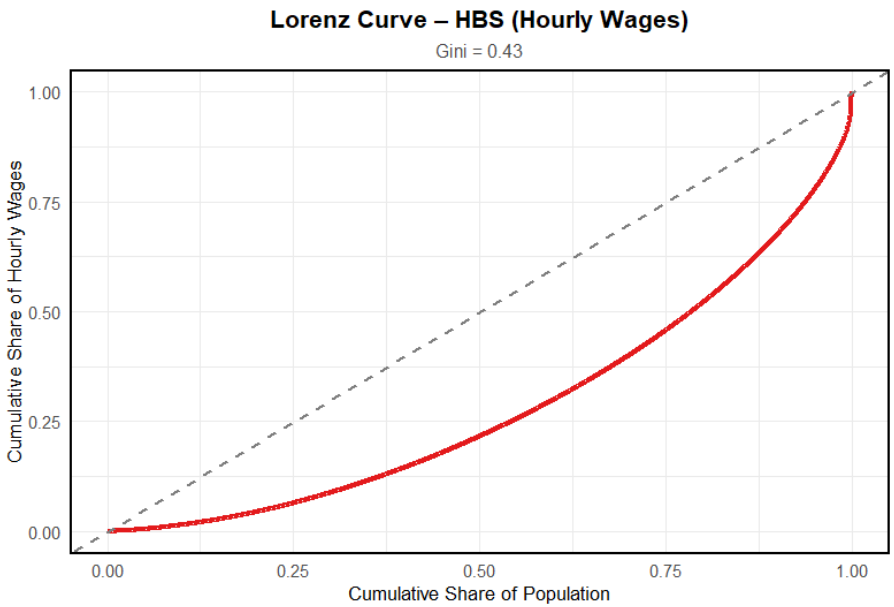
$$\log(\text{hourly wage}_i) = \beta_0 + \beta_1 \cdot \text{schooling}_i + \beta_2 \cdot \text{female}_i + \beta_3 \cdot \text{age}_i + \beta_4 \cdot \text{age}_i^2 + \sum_{r=1}^{R} \delta_r \cdot \text{Region}_{ir} + \varepsilon_i$$

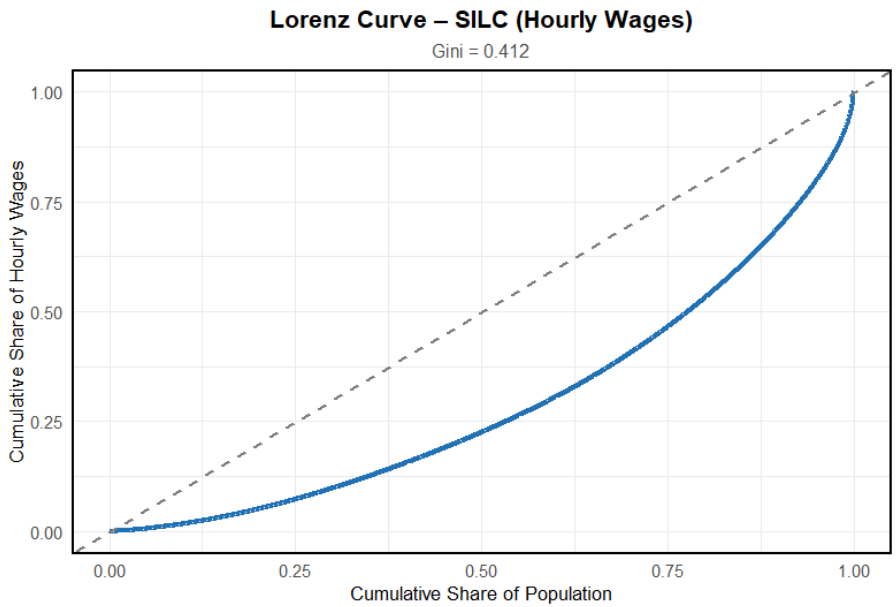
Determinants of Log Hourly Wage

	Dependent variable:	
	Log(Hourly Wage)	
	HBS 2022	SILC 2023
	(1)	(2)
Years of Schooling	1.8148***	1.2368***
	(0.1522)	(0.0917)
Female	0.0716***	0.0856***
	(0.0016)	(0.0011)
Age	−0.1821***	−0.1850***
	(0.0171)	(0.0108)
Age Squared	0.1134***	0.0806***
	(0.0072)	(0.0045)
age2	−0.0013***	−0.0009***
	(0.0001)	(0.0001)
TR2: Batı Marmara		−0.1318***
		(0.0239)
TR3: Ege		−0.2005***
		(0.0193)
TR4: Doğu Marmara		−0.0857***
		(0.0207)
TR5: Batı Anadolu		−0.1009***
		(0.0210)
TR6: Akdeniz		−0.2597***
		(0.0215)
TR7: Orta Anadolu		−0.1630***
		(0.0237)
TR8: Batı Karadeniz		−0.2905***
		(0.0211)
TR9: Doğu Karadeniz		−0.4784***
		(0.0288)
TRA: Kuzeydoğu Anadolu		−0.4137***
		(0.0238)
TRB: Ortadoğu Anadolu		−0.4226***
		(0.0233)
TRC: Güneydoğu Anadolu		−0.2613***
		(0.0207)
Observations	8,979	20,002
R^2	0.2265	0.2913
Adjusted R^2	0.2261	0.2908
Residual Std. Error	0.7108 (df = 8974)	0.6687 (df = 19986)
F Statistic	656.8368*** (df = 4; 8974)	547.6401*** (df = 15; 19986)

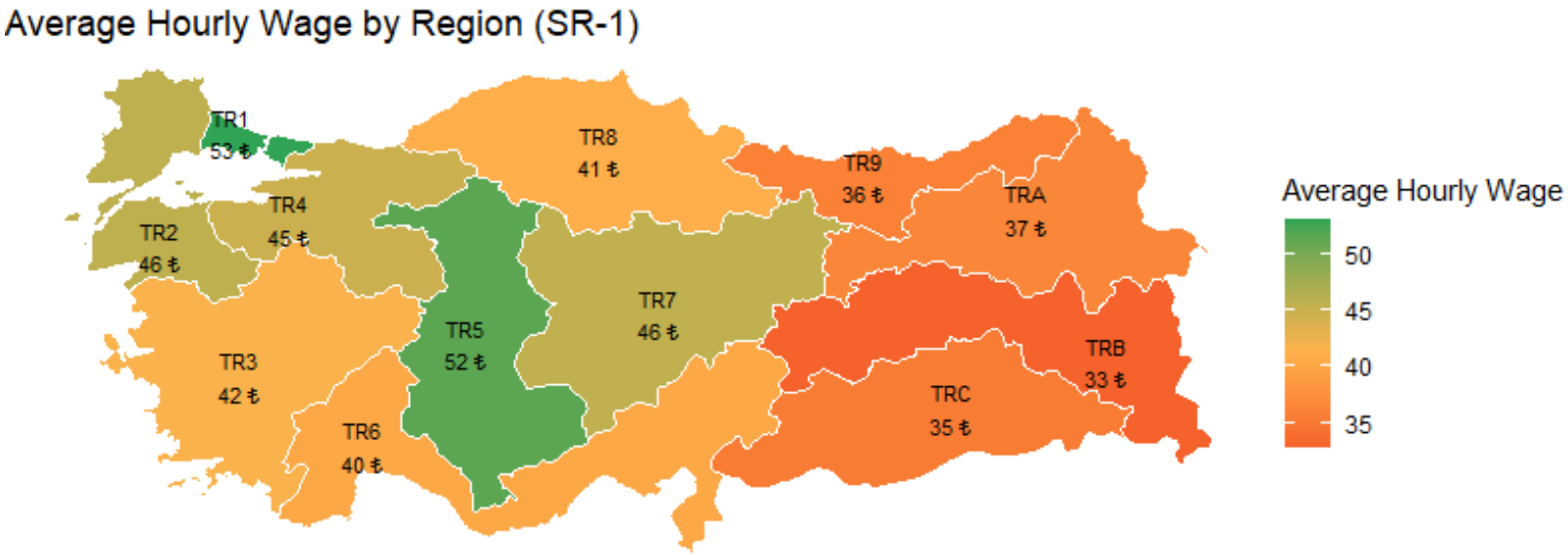
*p<0.1; **p<0.05; ***p<0.01. Standard errors in parentheses.

Lorenz Curves for HBS and SILC

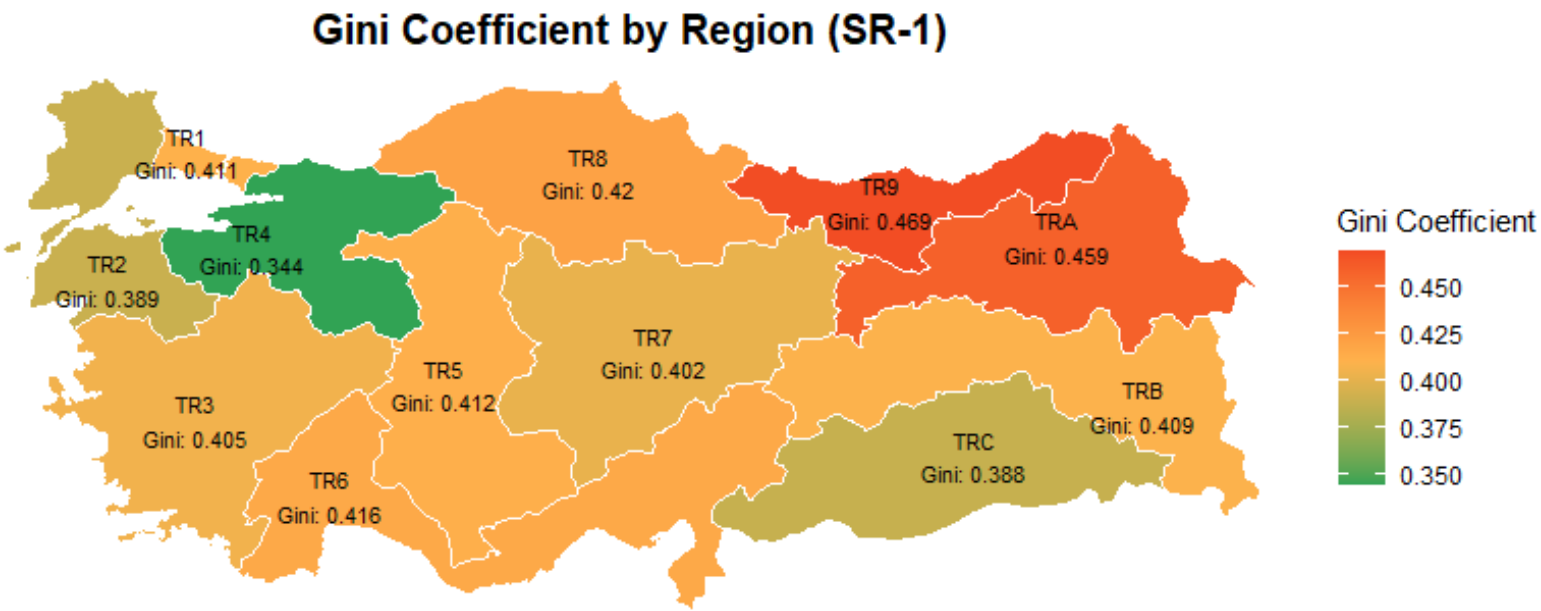




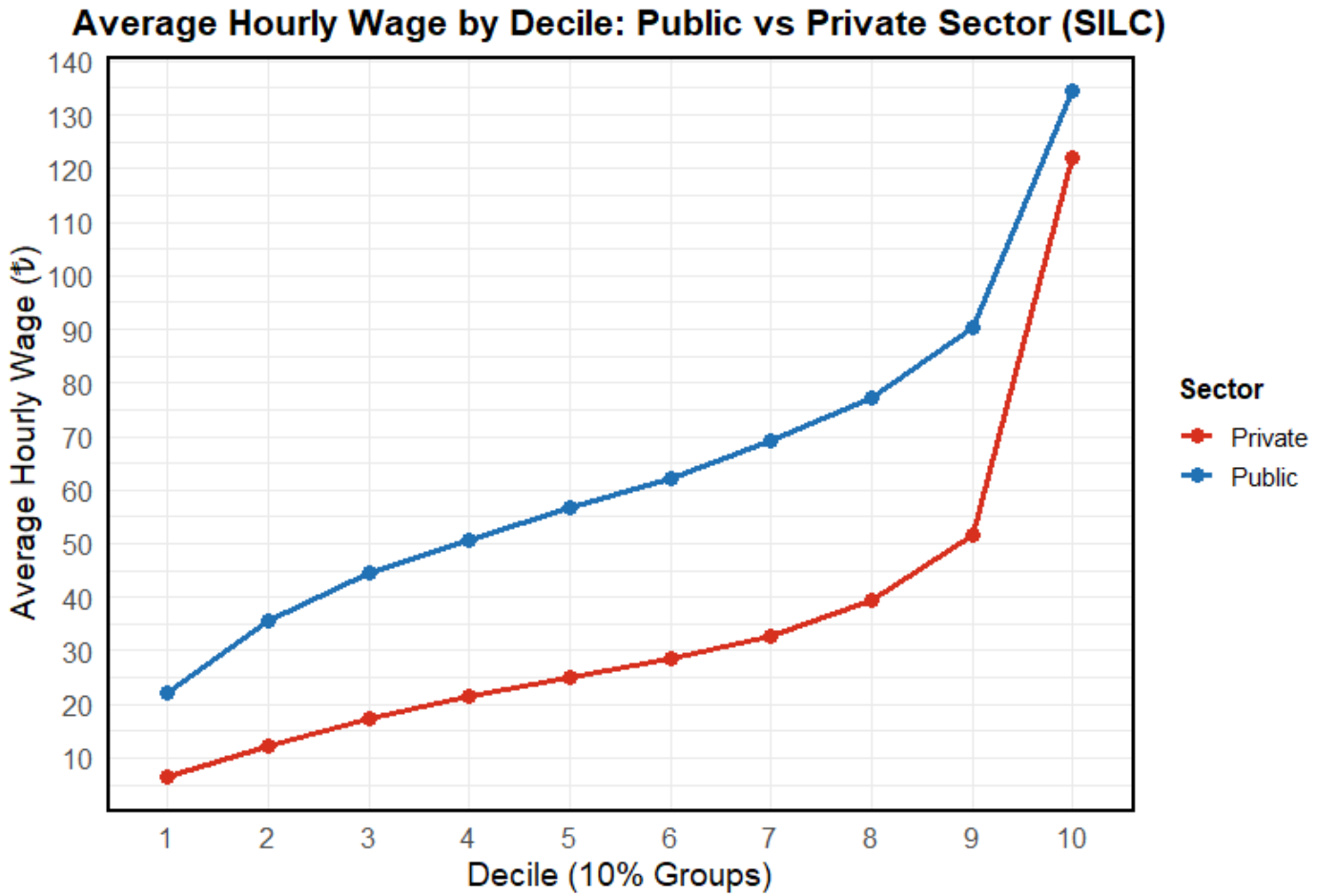
Average Hourly Wage



Regional Wage Inequality



Public vs Private Wage Inequality

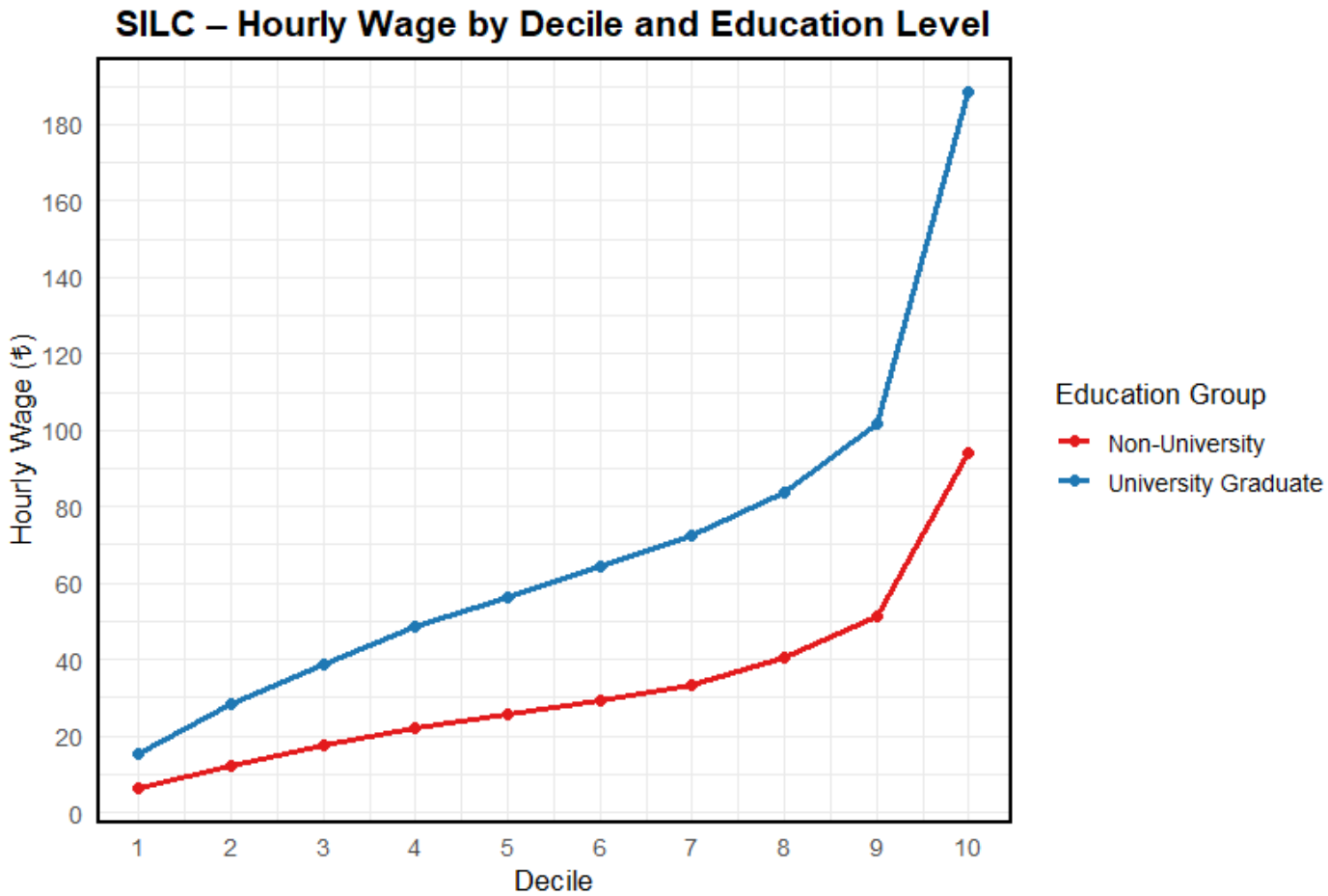


Gini Coefficients by Education Level

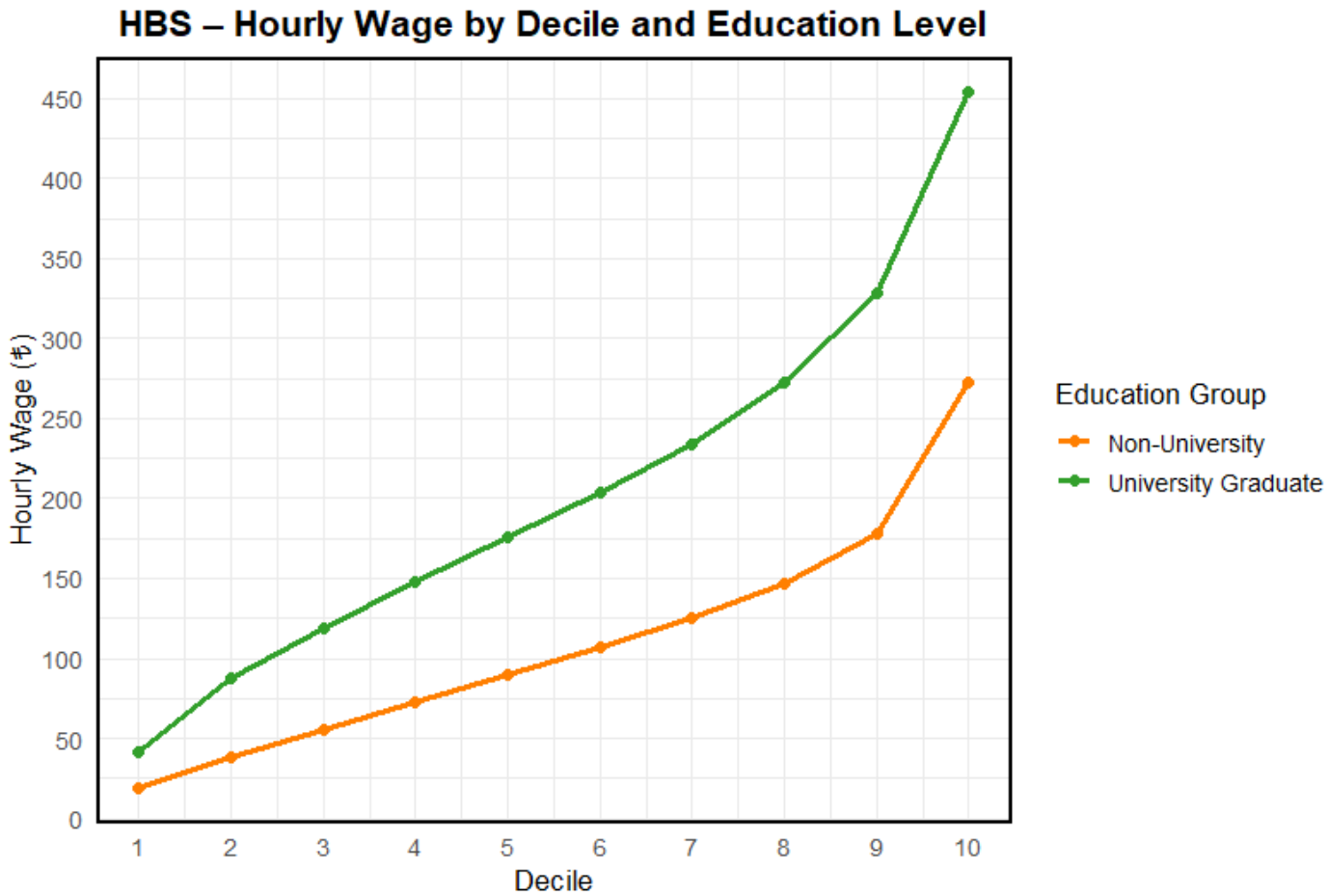
Group	Gini
SILC - University	0.3398
SILC - Non-University	0.3645
HBS - University	0.3124
HBS - Non-University	0.3523

- For SILC, higher education than "Faculty" are counted.
- For HBS, higher education than "Higher Educational Institutions for 2-3 years" are counted.

SILC Education Difference



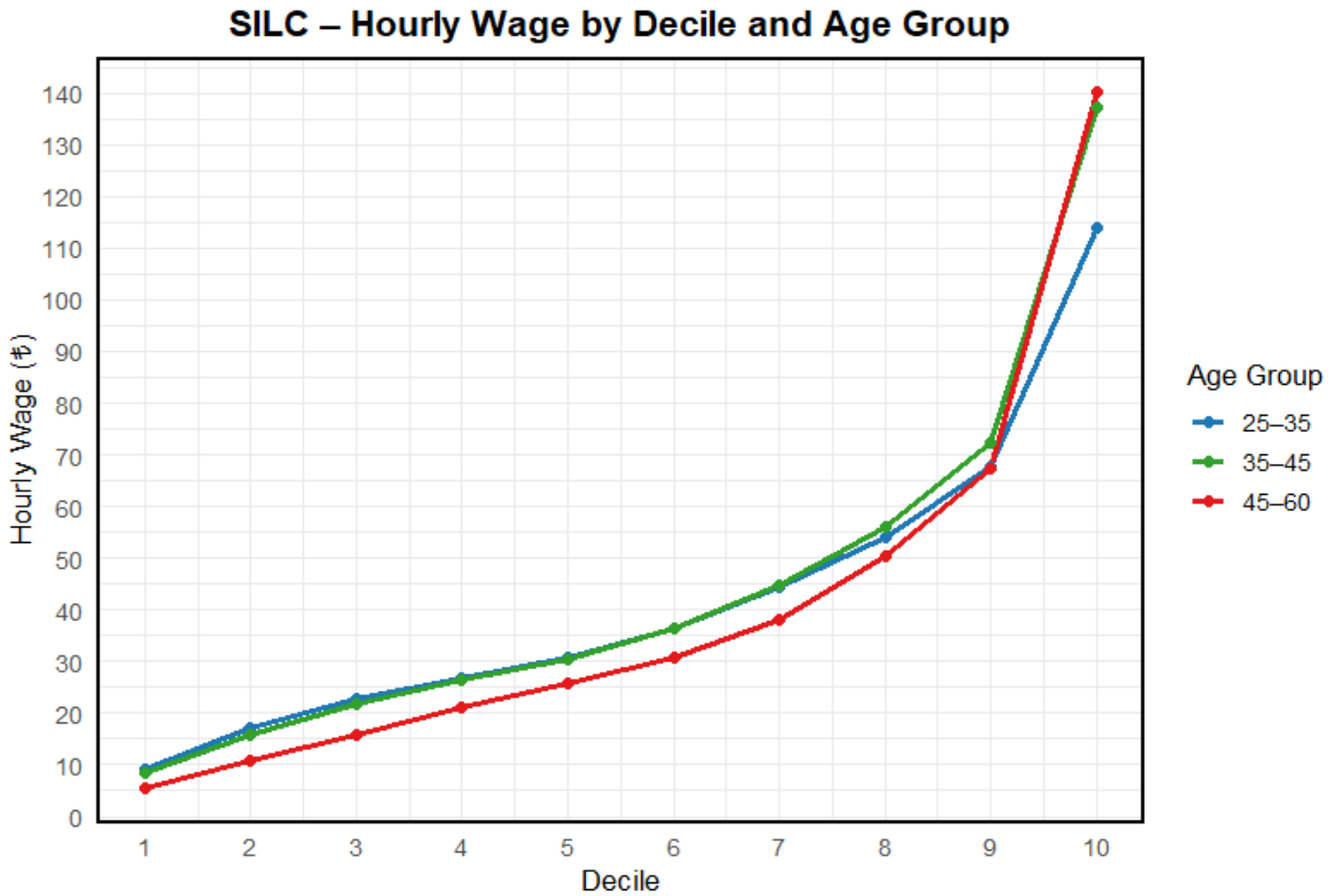
HBS Education Difference



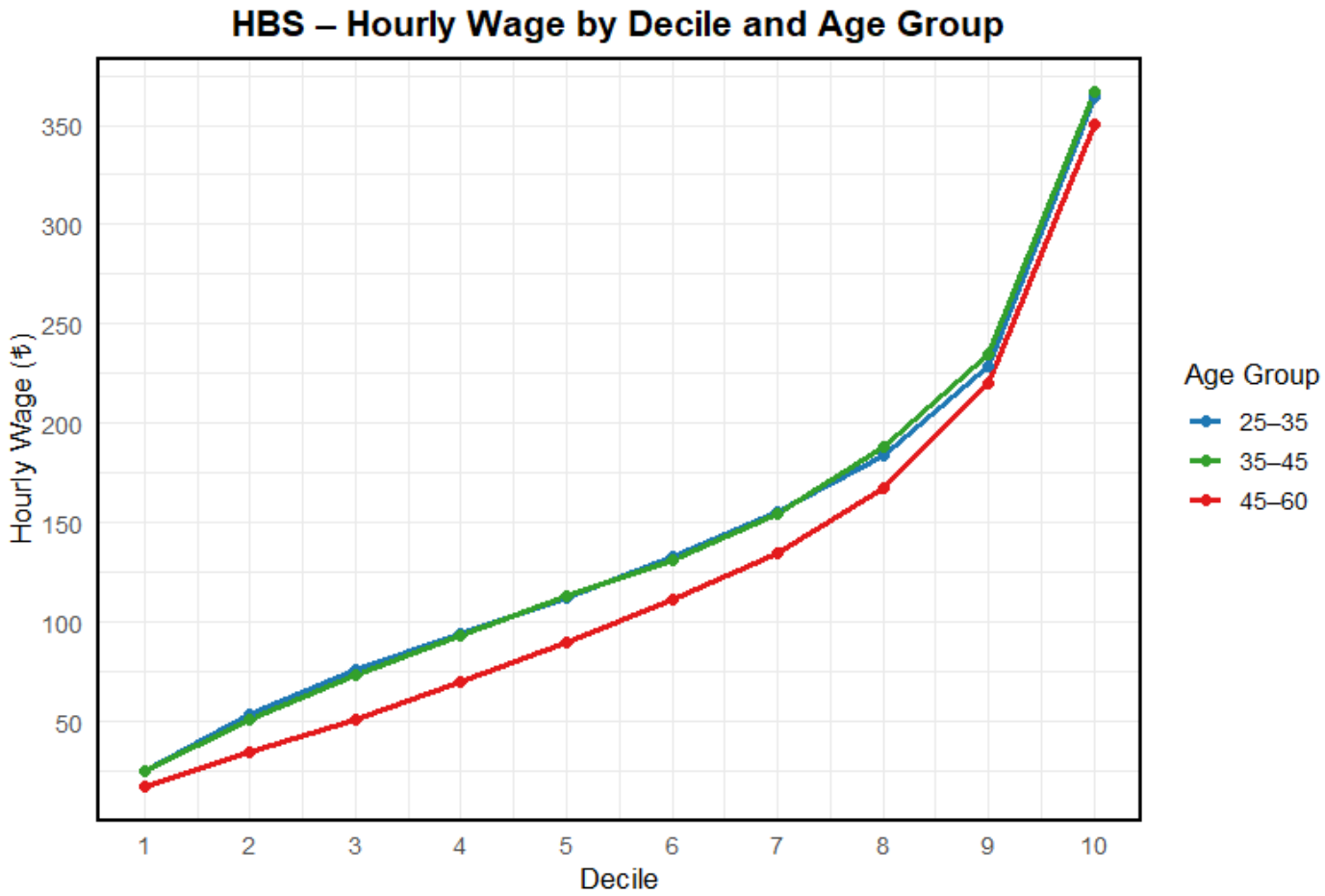
Gini Coefficients by Age Group

Age Group	SILC	HBS
25–35	0.3649	0.3575
35–45	0.4083	0.3648
45–60	0.4641	0.4142

SILC Age Group Wage Difference



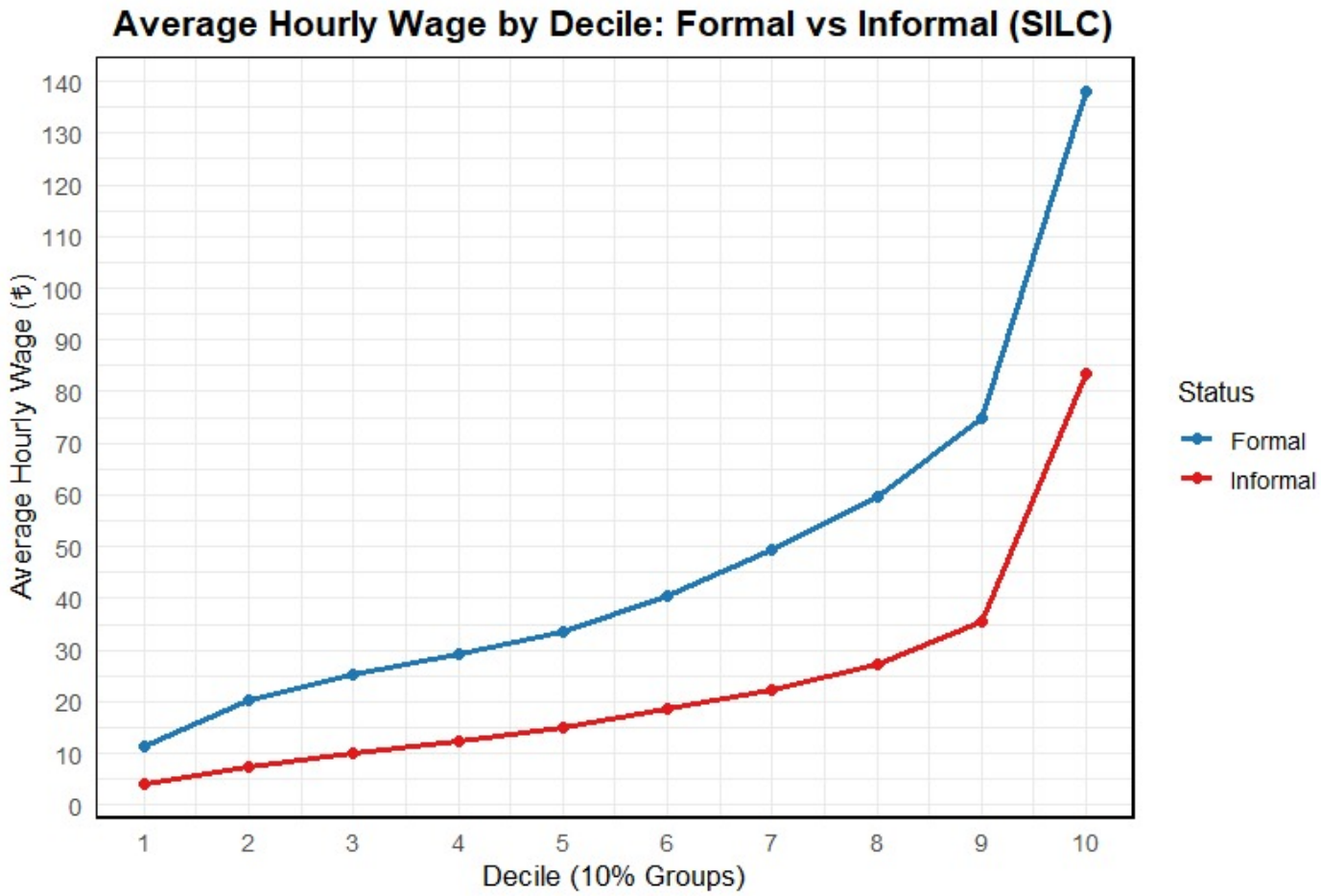
HBS Age Group Wage Difference



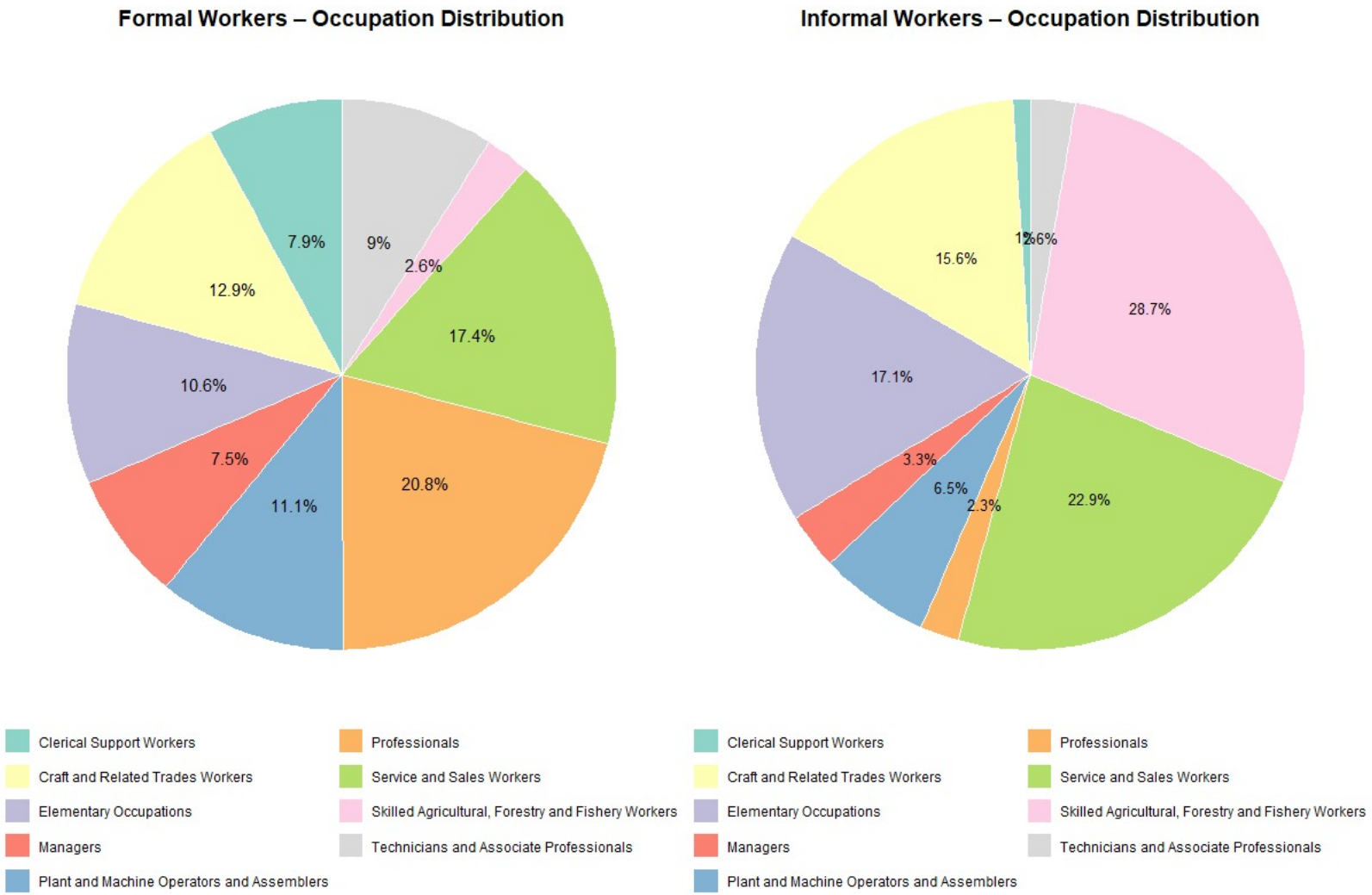
Summary Statistics by Employment Type (SILC)

Group	Mean	Median	Min	Max	Share
Formal	48.20	36.54	1.38	1079.06	77.9%
Informal	23.59	16.48	1.44	807.69	22.1%

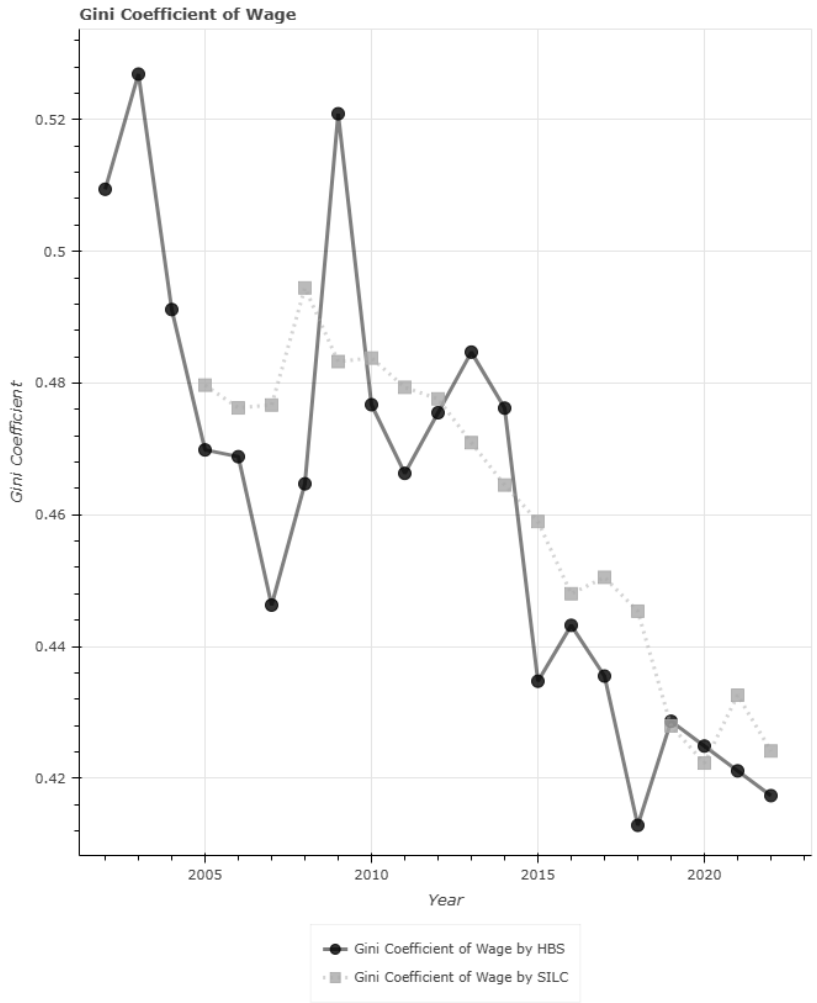
Formal - Informal Difference (SILC)



Formal - Informal Sector Differences (SILC)



From Past To Now Wage Inequality in Turkey



Source: https://web.bogazici.edu.tr/torul/csfm/

Note: Approximately same with our estimations for 2022.

- Comparative analysis of Turkey's wage inequality
  - Different studies as a reference for other countries' inequality measures.
  - Torul's findings for the previous years might be beneficial due to similar methodology and almost the same results for 2022 data.
  - Note that all premia have declining trend over 2002-2022 according to Tamkoç and Torul (2020)

Wage Inequality across Countries

	Gini	Var(Log Wages)	Edu. Premium	Gender Premium	Exp. Premium	P90/P50	P50/P10	Var(Residuals)
SILC	0.4120	0.6304	1.6463	1.0461	1.0509	2.5454	2.8448	0.4605
HBS	0.4304	0.6983	1.5623	1.0050	0.9175	2.3535	2.4871	0.5058

Table: Our estimations for 2022 data

Table 3. Wage Inequality and Wage Premia

Country	Level in year 2000				Change				
	Var. log w	College premium	Exp. premium	Gender premium	College premium	Exp. premium	Gender premium	Var. log w	Period
Canada	0.45	1.80	1.32	1.33	0.22	0.31	-0.11	0.17	1978-2006
Germany	0.27	1.38	1.27	1.28	-0.08	0.22	-0.15	0.05	1983-2003
Italy	0.17	1.51	1.34	1.03	-0.08	0.11	-0.05	0.03	1987-2006
Mexico	0.62	1.88	1.23	1.21	0.40	0.22	-0.06	0.04	1989-2002
Russia	0.77*	1.50	1.05*	1.49	-0.06	0.05*	-0.07	-0.13*	1998-2005
Spain(a)	0.23	1.48	1.43	1.16	-0.33	0.07	-0.21	-0.18	1985-1996
Sweden(b)	0.18	1.61	1.20	1.22	0.14	-0.02	-0.05	-0.09	1990-2001
UK	0.33	1.62*	1.25*	1.32	0.12*	0.20*	-0.21	0.10	1978-2005
USA	0.44*	1.80*	1.38*	1.36	0.40*	0.28*	-0.25*	0.21*	1980-2006
Average	**0.38**	**1.62**	**1.27**	**1.27**	**0.11**	**0.17**	**-0.10**	**0.04**	

A * indicates the statistic is from data on males only. Wage premia and wage dispersion for women are typically smaller.

(a) Data on changes refer to after-tax annual earnings

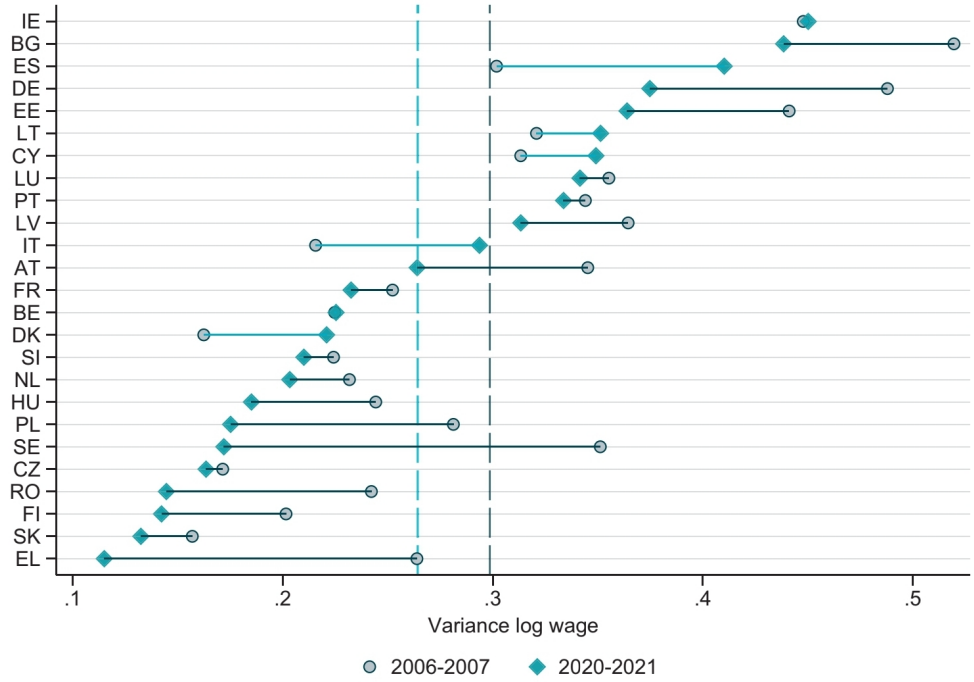
(b) Data on levels is for 1992

Source: Krueger et al. (2010) using RED guidelines.

Europe Case

"Wage inequality is declining overall, with some exceptions" demonstrated by Zwysen (2024).

Recall that variance of log wage is 0.6304 and 0.6983 in Turkey according to our estimations through SILC and HBS, respectively.[2]



Source: Zwysen (2024) via EU-SILC.

[2] Note that methodology may be different

Review of Wage Inequality in Turkey compared to Other Countries

- Turkey's wage inequality seems higher than developed countries.
- Gender premium seems lower than of those, maybe due to low female share of wage earners.

Review of the Literature and Factors' Impact

- **Real Minimum Wage Growth**
- **GDP Growth (countercyclicality)**
- **Formal Employment**

Impact of Real Minimum Wage

Case of Brazil shown by Engbom and Moser (2022, AER).
They show real minimum wage growth accounts for $\sim 45\%$ of the decline in wage inequality in Brazil while no significant effect on unemployment.

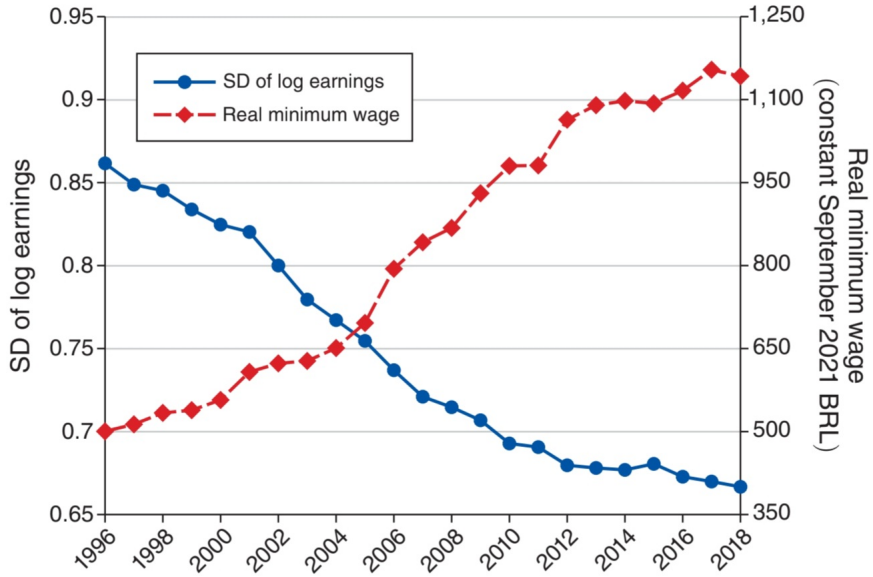


FIGURE 2. EVOLUTION OF WAGE INEQUALITY AND THE REAL MINIMUM WAGE

Notes: Statistics are for males of age 18–54. Real minimum wage is the annual mean of the monthly time series. The correlation between the two time series is −0.973. SD is standard deviation.

Sources: RAIS (1996–2018); IPEA (1996–2018)

Coincides with the results in Turkey that is demonstrated by Bakis and Polat (2021) and Tamkoç and Torul (2020)

Impact of Real Minimum Wage (Cont'd)

- Due to the high rate of minimum wage earners[3] and high increase in real minimum wage as compared to the increase in average real wages calculated by Tamkoç and Torul (2020) from HBS and SILC data.
- High inflation and decline in the real minimum wage in 2025 may have reversed the some fraction of the reduction in wage inequality through 2002-2022.

[3] Around 40% according to EuroStat

Change in Real Minimum Wage

Year	Annual CPI (2002=100%)	Monthly Minimum Wage Net (Real)
2002	100.00%	173.91
2003	121.60%	185.85
2004	132.10%	235.17
2005	142.90%	245.03
2006	156.60%	242.95
2007	170.30%	241.39
2008	188.10%	261.78
2009	199.80%	268.67
2010	216.90%	271.02
2011	231.00%	278.98
2012	251.50%	286.47
2013	270.36%	291.59
2014	294.30%	295.11
2015	316.85%	307.66
2016	341.15%	381.35
2017	379.57%	369.91
2018	441.57%	363.05
2019	508.58%	397.36
2020	571.03%	407.11
2021	682.93%	413.79
2022	1176.75%	414.44
2023	1805.59%	551.32
2024	2889.64%	588.38
2025[4]	4024.76%	549.20

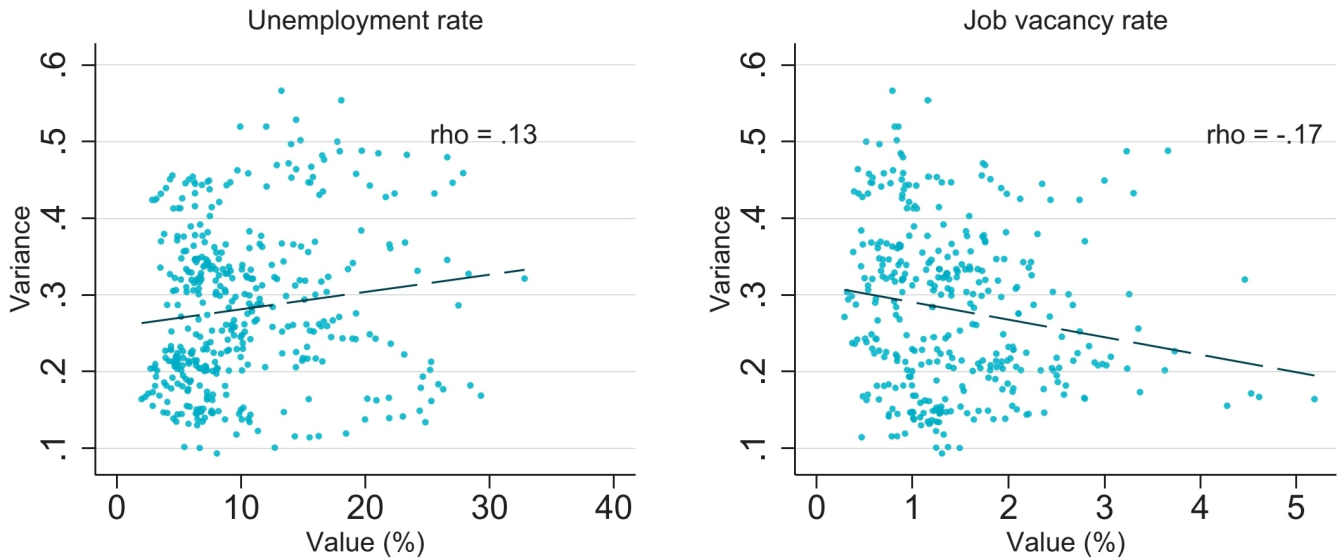
Table: Annual CPI and Real Net Minimum Wage (2002–2025). *Source:* TurkStat

- Growth rate of real monthly net minimum wage from 2002 to 2024 is 238.32%

[4] Calculated via the first 4 months' CPI

Labor Market

Zwysen argues that unemployment rate and job vacancy rate, which can be considered as a measure of demand for labor, have moderate correlations with variance of log wages.



Note: correlation between country-year level variance of log hourly real wages and contextual factors at country-year level, adjusted for breaks.
Source: EU-SILC 2007-2022, supplemented with external data.

Source: Zwysen (2024)

Countercyclicality



Figure: GDP growth vs. Gini Coefficient over years

Source: GDP growth from World Bank Data and Gini Coefficients from Tamkoç and Torul (2020)*(updated version)* due to similarity to our findings.

Decrease in Informal Employment

It is argued that most informal workers are at the bottom of the wage distribution by Tamkoç and Torul (2020). Therefore, wage inequality decreases mechanically as informal employment rate decreases.

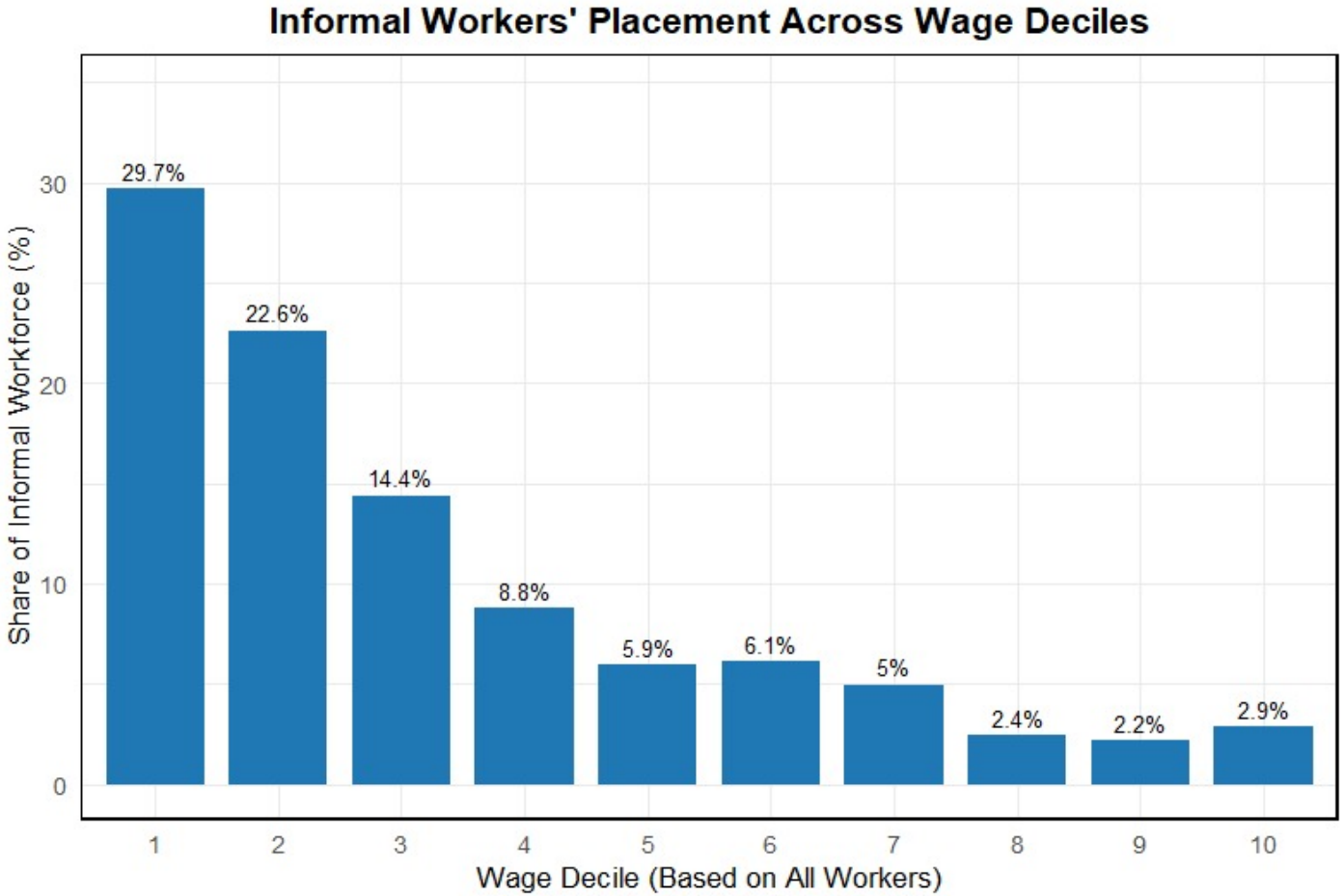
Table 1. Some characteristics of wage earners in Turkey

	Female (%)	Less than secondary share (%)	Years of schooling	Informal share (%)	Firm specific experience for private sector*	Small firm (≤ 10 workers) share for LTS** workers	Large firm (≥ 50 workers) share for ST*** workers
2002	21.1	56.7	8.7	29.8	5.5	47.9	49.7
2003	20.8	54.8	8.9	29.7	5.8	49.2	49.5
2004	21.0	56.9	8.6	31.8	5.5	47.5	47.0
2005	21.3	54.7	8.9	30.6	5.1	45.8	46.8
2006	22.0	53.4	9.0	29.5	4.7	45.2	46.4
2007	22.2	52.4	9.1	27.2	4.7	44.8	45.7
2008	22.8	51.1	9.3	23.8	4.4	42.9	45.6
2009	23.3	50.1	9.4	22.9	4.3	43.9	45.8
2010	23.3	50.8	9.4	22.3	4.1	42.7	45.7
2011	24.0	50.5	9.5	21.9	4.0	43.5	44.6
2012	25.2	48.6	9.6	18.9	4.0	43.8	45.1
2013	26.3	48.1	9.7	17.2	4.0	43.4	45.4
2014	27.0	48.6	9.7	16.2	3.9	45.0	46.7
2015	28.0	47.5	9.9	15.3	3.9	44.8	46.8
2016	28.9	46.0	10.0	14.6	4.1	43.7	47.6
2017	29.4	45.2	10.1	14.7	4.1	44.1	48.5
2018	30.4	44.6	10.1	14.7	4.1	43.0	49.6
2019	31.2	42.7	10.3	14.5	4.4	42.6	50.0

Source: TurkStat, HLFS 2002-19, Only positive wage earners are included in the sample. *Tenure years in current job, ** LTS stands for less than secondary education level, ***ST stands for secondary or tertiary education level

Source: Bakis and Polat (2021)

Note: Our estimation for the informal share of wage earners is $\sim 22\%$, using the registration status of Social Security Administration as a proxy in the SILC sample.



Source: Our estimations via SILC (2023)

Gender Factor

- Even though our estimation for gender premium in 2022 is approximately 1, which can be considered as no wage gap between female and male earners, there might be sample selection bias because we restricted the data for positive wage earners.

Gender	**Ratio**
SILC - Female	26.9%
HBS - Female	26.4%

Thank You!

Any question?

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